

Lecture 11

Introduction to Computer Network

1

Topics Covered (Chapter 12)

- Open System Interconnection (OSI)
 - Layer 1, Layer 2, etc.
- TCP/IP Network Model
 - Layer 1, Layer 2, etc.
- IP

2

2

1

Network Model

- Network models use layers to describe networks
- Each layer describes the services provided to the layer above it and those required from the layer below it
- It also describes the format of exchanges between peer layers on different network hosts
- Because the layers “stack” on top of one another, we often refer to network protocol “stacks” when we talk about the implementation

3

OSI Reference Model

The most well-known network model is the OSI (Open Systems Interconnect) Reference Model defined and maintained by the Organization for International Standardization (ISO). It consists of seven layers, numbered from the bottom (closest the network) to the top (closest the user).

4

OSI Reference Model

- Layer 1 – The Physical Layer
- Defines the type of media to be used
- Defines representation of data on the medium
- Is a '0' "high" or "low", "on" or "off"?
- What order are bits transmitted (if serial)?

Layer 1 – Physical

5

5

OSI Reference Model

- Layer 2 – The Data Link Layer
 - Defines "right to transmit" rules
 - Provides directly-connected host-to-host data transfer
 - Defines higher-level structure of data (frames)
 - Defines "physical" address structure for hosts

Layer 2 – Data Link

Layer 1 – Physical

6

6

OSI Reference Model

- Layer 3 – The Network Layer
 - Provides end-host-to-end-host data transfer across (potentially) multiple data links
 - Defines higher-level structure of data (packets)
 - Defines “abstract” address structure for hosts

Layer 3 – Network

Layer 2 – Data Link

Layer 1 – Physical

7

OSI Reference Model

- Layer 4 – The Transport Layer
 - Provides process-to-process data transfer
 - May provide for reliable data transfer
 - Defines higher-level structure for data (datagrams, streams, etc.)
 - Defines “port” addresses for services (processes)

Layer 4 – Transport

Layer 3 – Network

Layer 2 – Data Link

Layer 1 – Physical

8

OSI Reference Model

- Layer 5 – The Session Layer
 - Provides a logically persistent connection between processes
 - May involve user or host authentication (login), transaction encapsulation (for database access), etc.

Layer 5 – Session

Layer 4 – Transport

Layer 3 – Network

Layer 2 – Data Link

Layer 1 – Physical

9

OSI Reference Model

- Layer 6 – The Presentation Layer
 - Defines the network representation of data
 - Converts between the network and host representations of data (ASCII/EBCDIC, byte order, encryption, compression, etc.)

Layer 6 – Presentation

Layer 5 – Session

Layer 4 – Transport

Layer 3 – Network

Layer 2 – Data Link

Layer 1 – Physical

10

OSI Reference Model

- Layer 7 – The Application Layer
 - Provides a portal for the application to access the network
 - Describes the dialog between two applications communicating across the network.

Layer 7 – Application
Layer 6 – Presentation
Layer 5 – Session
Layer 4 – Transport
Layer 3 – Network
Layer 2 – Data Link
Layer 1 – Physical

11

11

TCP/IP Network (5-layered model)

- When TCP/IP was defined in the early days of the Internet, the OSI Reference Model had not been defined, so a different layering model was used
- It consists of 4 or 5 layers, and maps closely to the OSI Reference Model

Layer 5 – Application
Layer 4 – Transport
Layer 3 – Internetwork
Layer 2 – Link
Layer 1 – Physical

12

12

TCP/IP Network

- Layers 1 & 2 – The Physical and Link Layers
 - Provide physical communications between hosts within a network. Sometimes combined into a single “Link” layer.
 - Correspond to OSI layers 1 and 2

Layer 2 – Link

Layer 1 – Physical

13

13

TCP/IP Network

- Layer 3 – The Internetwork Layer
 - Provides all necessary components to move data between networks, including addressing, routing, etc.
 - Corresponds to OSI layer 3

Layer 3 – Internetwork

Layer 2 – Link

Layer 1 – Physical

14

14

TCP/IP Network

- Layer 4 – The Transport Layer
 - Provides everything necessary to move data between applications
 - Corresponds to OSI layer 4

Layer 4 – Transport

Layer 3 – Internetwork

Layer 2 – Link

Layer 1 – Physical

15

TCP/IP Network

- Layer 5 – The Application Layer
 - Provides everything specific to an application or a session
 - Corresponds to OSI layers 5 through 7

Layer 5 – Application

Layer 4 – Transport

Layer 3 – Internetwork

Layer 2 – Link

Layer 1 – Physical

16

Network Technologies –L1

- Layer 1 – The Physical Layer
 - Copper Cable
 - Coax – bulky, heavy, but relatively immune to noise
 - Twisted pair – thinner, lighter, cheaper, and okay about noise
 - Fiber Optic Cable
 - Multi-mode – good for relatively short distances (a couple of km) and moderate speeds (< 10 Gbps)
 - Single-mode – good for short and long distances (100+ km) and virtually any speed
 - Radio, Microwaves, Satellite, others
- None are inherently secure!

17

17

Network Technologies – L2

- Layer 2 – The Data Link Layer
 - Ethernet
 - Token Ring
 - FDDI
 - ATM
 - SLIP/PPP
 - Frame Relay
 - SONET
 - Waxed String

18

18

Network Technologies – L2

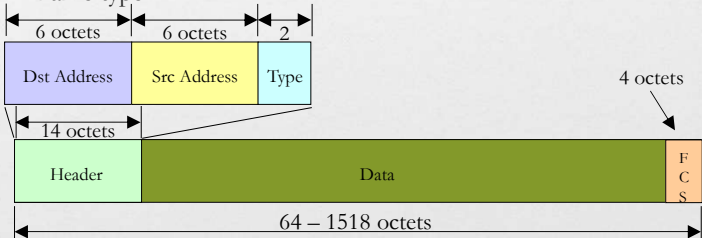
- Ethernet
 - Ethernet has a very simple transmission control protocol
 - Listen to the network
 - If someone is talking, wait your turn
 - If no one is talking, send your data
 - If someone starts talking while you are talking, stop talking and scream at them! ☺

19

19

Network Technologies – L2

- Ethernet transmissions occur in “frames” of 64 to 1518 octets in length
- The frame contains a header, data, and a checksum
- The header contains source and destination addresses and the frame type



20

20

Network Technologies – L2

- Early Ethernet devices had fairly basic functions
 - Transceivers physically connect hosts to the Ethernet coax cable
 - Repeaters amplify and repeat frames from one coax cable to another
 - Bridges *selectively* amplify and repeat frames from one coax cable to another

21

21

Network Technologies – L2

- Hubs provide a combination of convenience and flexibility
 - Coax hubs were a way to share a transceiver among multiple systems to avoid having to install so many. They used the same cable between a system and the hub as was used between a system and a transceiver
 - Twisted pair Ethernet hubs made it practical to provide network connections in office spaces
- Hub == Repeater!

22

22

Network Technologies – L2

- Switches provide for improved performance
 - Recall that bridges *selectively* forward frames from one network segment to another, while repeaters (hubs) *always* forward frames from one network segment to another
 - Switches are simply multi-port bridges implemented in hardware

23

23

Network Technologies – L2

- Finally, Wireless Access Points removed the need for cables between the network and the system
 - An access point is sort of a hub/switch hybrid
 - Individual systems have to share available bandwidth (radio waves)
 - Frames are selectively transmitted between network segments (wired/wireless)
 - Wireless systems may or may not be able to “hear” each other’s transmissions

24

24

Network Technologies – L3

- Layer 3 – The Network Layer
 - Layer 3 is where we connect layer 2 networks together
 - The layer 2 networks can be the same or different technologies
 - They can be the same or different speeds
 - Many of the concepts are the same as layer 2, only different

25

25

Network Technologies – L3

- IPv4 is the most successful layer 3 protocol ever developed
 - Hundreds of millions of systems
 - Every time zone
 - Every continent (yes, even Antarctica!)
- The basic protocol is unchanged since its inception in the early 80's

26

26

Network Technologies – L3

- IP Packets
 - Header
 - Addresses
 - Protocol
 - Control information
 - Payload (data)

32 bits

VER	IHL	TOS	Total Length	
Identification		Flags	Offset	
TTL	Protocol	Header Checksum		
Source Address				
Destination Address				
Options			Padding	
Payload				

27

27

Network Technologies – L3

- IP Addresses

172 . 24 . 57 . 18

10101100 00011000 00111001 00010010

28

28

Network Technologies – L3

- Network Masks
 - Network masks specify where the split between network and host portions is
 - A mask is a 32-bit quantity, just like an IP address, and can be represented the same way as a dotted “quad”

11111111 11111111 11111111 00000000
255 . 255 . 255 . 0

29

29

Network Technologies – L3

- Network Masks (cont'd)

172.24.57.18	→	10101100 00011000 00111001 00010010
255.255.255.0	→	11111111 11111111 11111111 00000000
172.24.57.0	←	10101100 00011000 00111001 00000000
172.156.62.23	→	10101100 10011100 00111110 00010111
255.255.255.240	→	11111111 11111111 11111111 11110000
172.156.62.16	←	10101100 10011100 00111110 00010000

30

30

Network Technologies – L3

- Network Masks (cont.)
- Knowing the network mask is important to understanding the structure of any IP address.
- But writing “172.24.57.18 with mask 255.255.255.0” is cumbersome.
- We could shorten it to “172.24.57.18/255.255.255.0”, but even that is longer than we need.

31

Network Technologies – L3

- Slash Notation

255.255.255.0

11111111 11111111 11111111 00000000

24 bits

255.255.255.240

11111111 11111111 11111111 11110000

28 bits

172.24.57.18/255.255.255.0

→

172.24.57.18/24

172.24.57.18/255.255.255.240

→

172.24.57.18/28

32

Network Technologies – L4

- TCP – The Transmission Control Protocol
 - Connection-oriented, Stream based, Reliable
 - In order, once and only once
 - “Heavy weight”
 - 3-way handshake
- Common uses:
 - Remote login (Telnet, RSH/Rlogin, SSH)
 - File Transfer (FTP, SCP, HTTP), Newer NFS

33

33

Network Technologies – L5+

- Layer 5+ - The Session, Presentation, and Application Layers
 - Telnet – Remote Login
 - File Transfer Protocol (FTP) – File Transfer
 - Simple Mail Transfer Protocol (SMTP) – Email
 - RSH/Rlogin – Remote Login
 - HyperText Transfer Protocol (HTTP) – World Wide Web
 - SSH/SCP – Secure Remote Login/File Transfer
 - **Note: Paper *ENEL802 Network Engineering* will discuss more on this topic.**

34

34